

Multiple Choice (10 marks)

1. You cool a blackbody to half its original temperature. How does its spectrum change?
 - (a) Power emitted is $1/16$ times as high; peak emission wavelength is $1/2$ as long.
 - (b) Power emitted is $1/4$ times as high; peak emission wavelength is 2 times longer.
 - (c) Power emitted is $1/4$ times as high; peak emission wavelength is $1/2$ as long.
 - (d) Power emitted is $1/16$ times as high; peak emission wavelength is 2 times longer.
2. A sand bag has a mass of 5 kg and weight 50 N on Earth. What is the mass and weight of this sand bag on a planet with half the gravity compared to Earth?
 - (a) Mass 5 kg weight 100 N
 - (b) Mass 5 kg weight 50 N
 - (c) Mass 5 kg weight 25 N
 - (d) Mass 10 kg weight 100 N
3. Our current best observations show that Pluto has
 - (a) one medium sized satellite and two small satellites.
 - (b) no satellites.
 - (c) one large satellite and three small satellites.
 - (d) one large satellite.
4. Mars has an atmosphere that is almost entirely carbon dioxide. Why isn't there a strong greenhouse effect keeping the planet warm?
 - (a) Mars does not have enough internal heat to drive the greenhouse effect
 - (b) Mars is too far from the sun for the greenhouse effect to work
 - (c) the greenhouse effect requires an ozone layer which Mars does not have
 - (d) the atmosphere on Mars is too thin to trap a significant amount of heat
5. The famous Drake equation attempts to answer the following question:
 - (a) Will the Sun become a black hole?
 - (b) Is the universe infinitely large?
 - (c) How old is the visible universe?
 - (d) Are we alone in the universe?

True / False (10 marks)

Answer True or False

6. True or False: A black body is an idealized physical object that reflects all electromagnetic radiation without absorption.
7. True or False: The Phoenix Mars Lander successfully landed on Mars on May 25, 2008, to investigate ice-rich soil in the polar region.
8. True or False: A scientific theory can be conclusively proven correct after a single successful experiment under controlled conditions.
9. True or False: The Large Magellanic Cloud is the closest planetary nebula to Earth.
10. True or False: Venus does not experience seasons because its rotation axis is nearly perpendicular to the plane of the Solar System.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the differences between sidereal and solar days, and critically analyze the statement that the Moon's circuit in our sky corresponds to one solar day.
12. Discuss the processes that led to the differentiation in material composition between the inner and outer planets in our solar system, focusing on the role of temperature in the nebula.

End of Paper